

Stakeholder Modification

Team: NYC Yellow-Taxi Fleet Allocation Agent

Released: two days before final presentations **Same dataset** — **no new data required**

Task 1 — Zone-Specific Overage Costs

The change in business reality

Your current model treats the cost of an idle cab as uniform across all zones within a time bucket — every idle cab costs the same regardless of where it sits. This is not realistic. A cab idling in a peripheral, low-density zone incurs a substantially higher opportunity cost than one idling in a dense corridor: the driver must travel farther to reach the next demand cluster, burning time and fuel before the next fare. Conversely, a cab staging at a high-throughput hub such as an airport expects to be re-engaged quickly, so its effective idle cost is lower.

We therefore replace the flat bucket-average overage cost Co_t with a **zone-specific overage cost**:

$$Co_z = \alpha_{b(z)} \cdot Co_t$$

where $b(z)$ is the borough of zone z and α_b is a stakeholder-supplied multiplier. The underage cost Cu_z (zone-level fare data) is unchanged. Everything else — fleet size, floors, caps, demand models — remains the same.

Overage cost multipliers

Use the borough-level multipliers below. The airport override applies to JFK (zone 132) and LGA (zone 138) specifically, regardless of borough.

Borough / zone group	Rationale	Multiplier α
Manhattan	Dense grid; short repositioning distance	1.00
Brooklyn	Semi-dense; moderate repositioning cost	1.25
Queens (non-airport)	Dispersed; higher repositioning cost	1.40
Bronx	Lower density; longer repositioning gaps	1.50
Staten Island	Isolated; very high repositioning cost	1.80
JFK (zone 132), LGA (zone 138)	High-throughput; rapid re-engagement	0.70

What to do

Update the cost pipeline so that Co_z is computed per zone using the multipliers above, and re-run the full allocation.

Produce a comparison across the two cost models for at least **three representative deployment slots** (e.g., Monday morning peak, Friday evening, Saturday midday):

Model	What it is
(A) Flat overage cost	current model — $\text{Co}_z = \text{Co}_t$ for all zones
(B) Zone-specific overage	$\text{Co}_z = \alpha_{b(z)} \cdot \text{Co}_t$ with supplied multipliers

For each model and slot report:

1. the fleet allocation vector q^* (summarized by borough totals);
2. realized newsvendor cost on the held-out test weeks;
3. the critical fractile $\tau_z = \text{Cu}_z / (\text{Cu}_z + \text{Co}_z)$ for the five highest-shift zones between (A) and (B).

Then answer, in plain language:

- Which boroughs gain fleet and which lose it under Model B, and does the direction match economic intuition?
- How do the airport zones behave? Does the lower overage cost there cause the agent to pre-position more or fewer cabs, and why?
- Does Model B produce a lower realized newsvendor cost than Model A on the test set, and by how much? If not, explain why the flat approximation was adequate.
- Your proposal states the agent can absorb new cost parameters without code restructuring. Confirm this, and describe what precisely had to change in the pipeline.

Cross-check requirement. Your solver suite already includes a three-way cross-check (SLSQP, water-filling, cvxpy). Run it under Model B and confirm the solvers still agree within your established tolerance. If they disagree, diagnose why.

Task 2 — Fleet Size as a Decision Variable

The change in business reality

Your current model fixes the fleet at 13,000 cabs — a constant passed in from TLC operations. TLC has now indicated that this is not a hard constraint. For any deployment slot the fleet size F can be adjusted: additional cabs can be brought on-line at a rental rate of **\$45 per cab per slot**, and cabs can be stood down to yield a cost saving of **\$20 per cab per slot** relative to the baseline. The decision of how many cabs to deploy is therefore jointly optimizable with the spatial allocation.

The extended formulation

Let F denote the total fleet size for a given slot and let $F_0 = 13,000$ be the baseline. The fleet adjustment cost is:

$$\text{cost}(F) = r (F - F_0)^+ - s (F_0 - F)^+$$

with $r = \$45$ per cab per slot (rental) and $s = \$20$ per cab per slot (stand-down saving). The joint objective becomes:

$$\min_{F, q \geq 0} \underbrace{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_z \left[\text{Cu}_z (D_{z,t} - q_z)^+ + \text{Co}_z (q_z - D_{z,t})^+ \right]}_{\text{newsvendor cost}} + \text{cost}(F)$$

subject to:

$$\sum_z q_z \leq F, \quad F \geq F_{\min}, \quad \ell_z \leq q_z \leq u_z \quad \forall z$$

where Co_z is the zone-specific cost from Task 1, $F_{\min}$ is a TLC-mandated minimum coverage floor (use the minimum F at which all per-zone floors ℓ_z are simultaneously satisfiable), and all other constraints are unchanged.

Note on asymmetry. The rental rate r and the stand-down saving s are different ($\$45 \neq \20). This asymmetry has consequences for how the agent behaves around the baseline fleet size. Address this in your explanation.

What to do

Extend your solver to treat F as a continuous decision variable jointly with q . Re-run across the same three deployment slots used in Task 1.

Produce a comparison across three fleet models:

Model	What it is
(A) Fixed fleet, flat cost	Task-1 Model A baseline
(B) Fixed fleet, zone cost	Task-1 Model B
(C) Elastic fleet, zone cost	Task-2: F optimized jointly with q

For each slot report:

1. the optimal fleet size F^* and the direction of adjustment from the 13,000 baseline;
2. the shadow price on the fleet capacity constraint $\sum_z q_z \leq F$ — this is the marginal value of one additional cab;
3. total cost (newsvendor + fleet adjustment) vs. Model B;
4. the break-even rental rate: the value of r at which TLC is indifferent between renting an additional cab and accepting the newsvendor loss.

Then answer, in plain language:

- Does the agent expand or contract the fleet, and does the answer vary by slot? What drives the difference?
- How does the $r \neq s$ asymmetry affect the agent's decision around the baseline fleet size?
- At the optimal F^* , is the fleet capacity constraint binding? What does the shadow price tell TLC about the value of its cab inventory?
- How much of the total cost reduction ($A \rightarrow C$) comes from better spatial allocation (Task 1) versus fleet right-sizing (Task 2)?

Deliverables

1. **Task 1:** updated cost pipeline producing zone-specific Co_z ; the (A) vs. (B) comparison table for the three slots; solver cross-check results under Model B.
2. **Task 2:** extended solver with F as a decision variable; the (A) / (B) / (C) comparison table with fleet sizes, costs, shadow prices, and break-even rental rates.
3. **Written explanation** covering the plain-language questions in Sections 1.3 and 2.3.